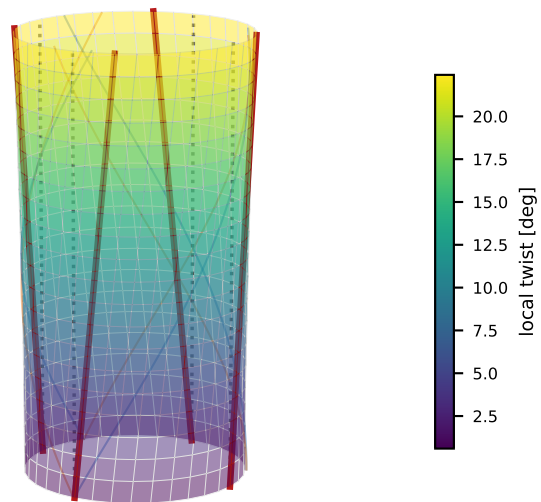
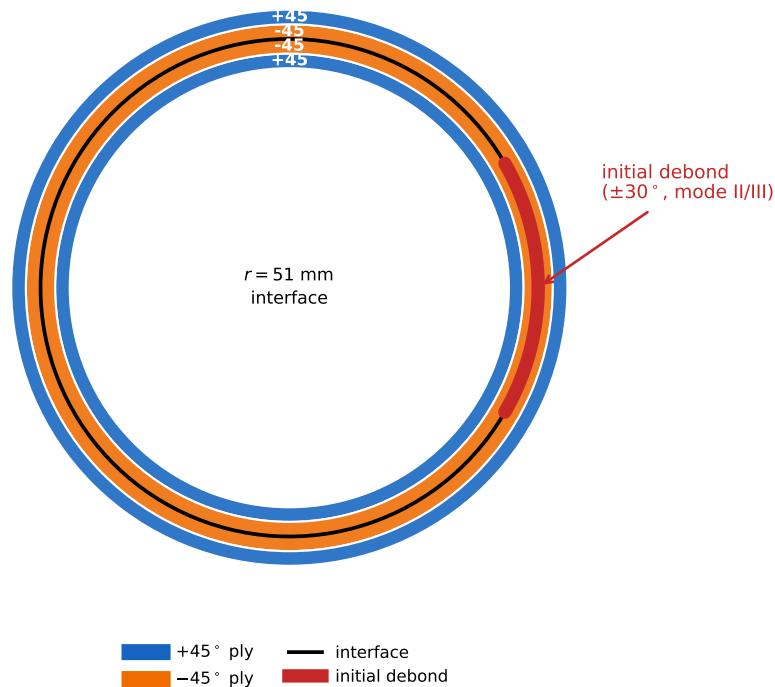


CFRP drive shaft under torsion: 3-D ± 45 laminate with interlaminar cohesive zones

(a) torsion: top rim $+3^\circ$ (deformation $\times 8$)
red=deformed generators (helical), dotted=undeformed;
faint blue/orange = ± 45 fibre paths



(b) wall cross-section: $[+45/-45 \parallel -45/+45]$
(radial scale $\times 6$; interface + initial debond)



(c) torque-rotation
(3D ± 45 + interlaminar CZM)

